Harvested Energy Prediction Technique for Solar-Powered Wireless Sensor Networks

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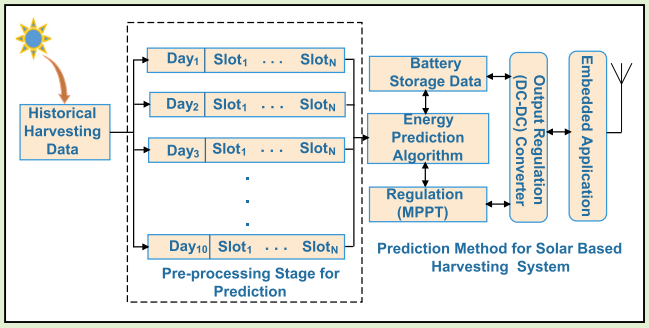
***Abstract*—Solar energy harvesting (EH) is one of the best promising approaches toward perpetual network operation, and it is implemented in various regions of interest (RoIs). However, saving external energy is the essential prime factor in any embedded sensor with finite storage capacity. Generally, the energy conversion rate of the solar system is too fast due to various environmental conditions. Besides, ambient resource energy is noncontrollable, and rechargeable battery only operates in outdoor ecological systems. Frequent environmental fluctuation in their prediction is imperative for initial energy control. Considering the challenges mentioned above, in this article, we propose a modified PROfile energy (Pro-energy) prediction technique to control unnecessary errors in solar-based harvesting systems related to the sensing devices, which estimates the most similar profile-based energy observation in previous time slots. Our proposed method uses prior energy measurements to show future energy status in the respective time slots. Experimental observations on various performance matrices validate that the modified Pro-energy prediction technique exhibits more promising and superior performance than existing EMWA, weather-conditioned moving average (WCMA), and Pro-energy methods.**



***Index Terms*—Energy management, prediction, renewable energy harvesting (EH), solar energy, wireless sensor networks (WSNs).**

Nomenclature

C	Harvested energy on the current day [W/m^2].
N	Total number of time-slots in a day.
D	Stored past day energy profile.
MAE	Mean absolute error [%].
E	Matrix to store multiple energy profiles in $D \times N$.
P	Identify the number of energy profiles.
WP	Weight profile within combine P.
H	Actual harvested energy in previous slots.
M_D	Calculated mean value of past day energy samples.
$\widehat{E}(d,n)$	Predicted energy information at the dth day of the nth slot.
E_L	Energy level of the battery [J].
$\rho(d,n)$	Expected error in $\widehat{E}(d,n)$ [%].
$H(d-1,n)$	Energy harvested in past day time slot [W/m^2].
$I_{\text{solar-cell}}$	Node fetches solar irradiance value.
α	Weighting factor value for prediction.
$E(d-1,n)$	Expected energy in past day time slot [W/m^2].

Manuscript received 1 August 2022; accepted 19 September 2022. Date of publication 28 September 2022; date of current version 14 April 2023. The associate editor coordinating the review of this article and approving it for publication was Prof. Rosario Morello. *(Corresponding author: Tarachand Amgoth.)*

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Digital Object Identifier 10.1109/JSEN.2022.3208730

I. Introduction

WIRELESS sensor networks (WSNs) include low-constrained energy sensor nodes, performing distinct task execution in real-time applications. These applications can range in various domains, such as smart home, border surveillance, environment system tracking, Internet of Things (IoT), and vehicle tracking, and so on [1], [2]. The sensing devices comprise complex practical components: the transceiver unit, power unit, sensing unit, and an analog-to-digital converter (ADC). Out of the four elements, the power unit is critical because there are so many tasks accomplished between the sensor node and the sink node using routing techniques [3]. The routing technique may be single-hop routing or multihop routing. The sensor network's energy rate consumption is highly rapid during the routing process. The primary sources

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of energy depletion in the sensor network have included overhearing, retransmission of the packet, inactive listing, and control packet overhead. Moreover, the battery capacity of the traditional sensor nodes is nonrechargeable. It is powered by minimum power sources due to the short-range recharge cycle. Usually, sensor nodes are deployed in unbearable weather conditions, thereby manually replacing the battery after a specific period is too tricky [4].

The role of the prediction approach in solar-based harvesting is extremely prime to preserve essential energy resources from the environment because all the network load operations depend on the battery's energy level. Therefore, we have divided the application part into two ways: mainly indoor and outdoor. There are many applications under indoor systems, such as green smart homes, industrial IoT (IIoT) devices, and home automation. On the other hand, the outdoor application is based on the green agriculture industry 4.0, green smart city, green IoT, and solar street lights.

A. Motivation

Over time, researchers have explained the limitation of energy resources by employing the energy-harvesting (EH) technique. The EH techniques can be of different types, such as solar, wind, mechanical, and thermal, and highly effective solutions in the sensor network with a lesser cost [5]. The renewable energy system provides unlimited energy resources and removes just a bottleneck problem of the energy resources in all the limited energy systems in WSNs. Besides, the harvesting system's general principle is to convert ambient resources into electricity, supplying the energy to the sensor nodes device [6]. Many components can be used in the EH system, namely rechargeable battery, super-capacitor, harvesting transducers, energy predictor, and power management unit (PMU) [7]. In the harvesting system, the purpose of the energy predictor is significant because an ambient energy resource is uncountable, and their prediction must provide effective energy management in WSNs [8]. Moreover, the solar and wind harvesting systems entirely depend on outdoor weather conditions, which means solar energy is based on a sunny day, and wind energy depends on air mass. The harvesting technique is a very dynamic system, and the harvesting rate constantly changes according to the weather condition; therefore, we can manage the boundless energy using prediction techniques [9]. The practical use of the EH system is justified using the erratic behavior of the energy resources. It implies the amount of harvested energy and energy rate over the daily time slot. Therefore, the prediction technique has been developed to adjust the power needed for sensor devices from ambient energy resources [10].

B. Related Work

Over time, several research initiatives have been done related to the prediction techniques in EH-WSNs. For example, Ashraf *et al.* [11] have addressed a new prediction method called the accurate solar irradiance prediction model (ASIM). It works on short- and long-term data using real datasets. The proposed method uses a nonlinear model to show robust throughput in the sensor network. Yang *et al.* [12] have designed a solar-powered WSN (SP-WSN) to handle monitoring applications. The device is built for self-protection and self-healing, thereby it can handle solar prediction errors with minimum risk of network failure. Hsu *et al.* [13] have demonstrated an operational duty cycle to adjust power management in the sensor network. Reinforcement learning identifies better power management in networks through various states. Cammarano *et al.* [14] have introduced PROfile energy (Pro-energy) variable length timeslot (VLT) to solve multisource EH prediction problems. This method works with different datasets, such as wind and solar data. In another work, Kosunalp [15] have elaborated a novel solar energy prediction technique and proposed Q-learning-based solar energy prediction.

Similarly, Dehwah *et al.* [16] have incorporated an energy management scheme with a solar-powered EH system. Two algorithms were used in the proposed work: mixed integer linear programming (MILP) and dynamic programming. Here, Ganjewar *et al.* [17] introduced the transmission technique using hierarchical least-mean square (HLMS). The proposed method can predict sensed data in the sensor network, which means sensor nodes to the base station. They have shown neutral power management using prediction techniques, which is more effective in the sensor networks [18]. One focus of the proposed algorithm is to manage the energy budget of the sensor node. Gorlatova *et al.* [19] have designed a low-power EH device to control the uncontrollable energy resources. The work is based on the long-term indoor solar radiant energy measurement, and the author primarily considers the battery energy level. Herrería-Alonso *et al.* [20] have identified different angle variations for solar EH prediction systems in EH-WSNs. The factors can be possible in different ways, just as solar altitude angle of light and energy prediction. Zou *et al.* [21] have proposed a cluster-based routing technique using prediction technique optimization in EH-WSNs. The performance metric is justified for elegant prediction, especially harvest prediction and energy control, time correlation prediction, and an extended Kalman filter. Literature shows that four essential components are needed for solar-based prediction, namely standard dataset, efficient solar hardware kit (protect the device due to the high rate fluctuation of solar energy), prediction module (short- and medium-term), and self-intelligence (for error minimization). But, most of the previous research works did not use all components.

C. Contribution

Motivated by the above-mentioned challenges in mind, we have illustrated a modified Pro-energy prediction procedure to resolve the error problem in EH-WSNs. Pro-energy depends on the most similar profile in the respective time slot of day to identify better energy observation between harvested and estimated energy. Our proposed method shows an accurate estimation of the amount of harvested energy by the solar panels to use shortly in the following days. Therefore, management of the energy resources can be productive in the deployed location of the sensor nodes in EH-WSNs. We have summarized our contributions as follows.

1) This article is based on the energy prediction model, and it is a primary process for energy management, where

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energy prediction pattern leverages earlier energy analyses for accurate estimations of future energy availability.

2) We propose an effective prediction method called modified Pro-energy prediction, which only depends on the outdoor solar-harvesting prediction model.
3) The algorithm worked based on the most similar profile and showed short-term prediction to estimate an accurate prediction ratio in the given time slots.
4) The modified Pro-energy method is more accurate and robust based on minor prediction errors. As a result, the battery's energy level would also increase.
5) Work performance is identified as better parallel error reduction of previous time slots during the process, thereby beating other techniques.

The remaining sections of this article are organized as follows. Section II showed the problem statement and solar EH model. We have described the work proposal in Section III, and it is discussed various types of result analysis in Section IV. Finally, Section V is the conclusion part of the work.

II. Preliminaries

This section presents essential concepts for prediction techniques, problem statements, and solar harvesting prediction models. Nomenclature contains all variables used in the system model.

A. Existing Energy Prediction Techniques

In the literature, Kosunalp [22] have proposed some previous prediction techniques, especially exponential weighted moving average (EWMA) and weather-conditioned moving average (WCMA). EMWA has shown last-day prediction energy concerning different time slots of the day. However, this method calculated historical information on the EH rate. Equation (1) has identified the last amount of harvesting energy (H) and estimated energy (E) by the EMWA using a weighting factor $0 < \alpha < 1$, if the value of α is higher, then less chance the harvesting rate of energy vice versa. In (1), d represents the present-day within the nth slots

$$E(d, n) = \alpha E(d-1, n) + (1-\alpha)H(d-1, n). \tag{1}$$

The different values of α have been calculated to show various seasonal changes related to the weather condition in the harvesting system. Moreover, the value of α is crucial in the prediction techniques. The EMWA prediction technique has drawbacks, such as no storage matrix of the last harvested energy information. More prediction errors are possible due to the immediate change in the weather condition, just as sunny to the cloud and vice versa.

Moreover, the WCMA is an extension of the EWMA prediction technique. The WCMA method stores previous harvesting value using matrix $E(i, j)$, with the ith day on the jth slot. However, the WCMA method collects harvested data of the relevant slots of the previous day. The value of α is a weight factor same as the EMWA technique, and H is actual harvested energy in the last time slots, then (2) as follows:

$$E(d, n) = \alpha H + (1-\alpha)M_D(d, n)\mathrm{GAP}_k \tag{2}$$

where M_D is the mean value of the previous day. The mean value of past day energy observed D as

$$M_D(d, n) = \frac{\sum_{i=d-D}^{d-1} E(i, n)}{D}. \tag{3}$$

The main contribution of the WCMA is the GAP_k value factor. This method was calculated to measure the solar harvesting rate condition of the present day to see previous day conditions. However, to the compute GAP_k factor, we first identify a vector $V = [v_1, v_2, \ldots, v_K]$ with the past K slot elements. Each sample of harvested energy denotes the ratio of the harvesting energy to the average mean value. However, (4) identifies that the current-day value is larger than the calculated mean value, showing a sunny day, otherwise a cloudy day

$$V_k = \frac{E(d, n-K+k)}{M_D(d, n-K+k)}. \tag{4}$$

However, it is vital to get more weight in the closet sample of the day on time. The WCMA technique has identified another vector $F = [F_1, F_2, F_3, \ldots, F_K]$, where each vector is defined according to the distance to the actual slot

$$F_k = \frac{k}{K}. \tag{5}$$

Finally, the weight factor GAP_k has computed as

$$\mathrm{GAP}_k = \frac{V.F}{\sum F} = \frac{v_1 f_1 + v_2 f_2 +, \ldots, + v_k f_k}{f_1 + f_2 +, \ldots, + f_k}. \tag{6}$$

The main drawback of the WCMA prediction technique is the high computational time. Generally, this technique takes too much time to calculate the GAP_k value on different days for a particular sample in the various slots. Another issue may be possibly related to the value of α. If α is a constant value, it creates the problem because α identifies different seasonal changes in the harvesting system. Moreover, it may be a possible error due to more weight on recent slots.

B. Problem Statement

One of the most challenging issues in the EH system is energy prediction because the harvesting rate entirely depends on different types of weather conditions. However, the solar-based prediction model is a critical justification for how much energy will come in the next day in varying time slots. Generally, solar- and wind-harvesting systems depend on the outdoor system, which means high solar light intensity and the mass number of wind. However, this technique is the initial process of controlling ambient resources management; if the prediction ratio is accurate, the battery's residual energy will be high. The energy predictor device directly connects with the solar-harvesting device's rechargeable battery in EH-WSNs. The sensor node is often scattered in the unreachable location, and data are regularly transmitted at the base station. Therefore, maintaining the prediction technique's preservation is a fundamental issue for the EH-WSN. Another problem is based on the hardware quality of solar-harvesting devices. Generally, solar light's energy conversion

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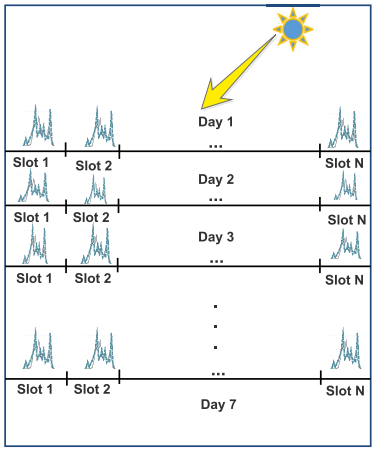

Fig. 1. Diagram of solar harvesting within different time slots.

rate is high in the harvesting system; therefore, some troubles with the components would be affected. For example, it may create leakage and chemical reaction with storage classes (rechargeable battery/supercapacitor). It is also possible due to the storage capacity versus the amount of the harvested energy. An accurate irradiance measurement issue comes with maximum power point tracking (MPPT) device in different time slots. Therefore, the role of an energy predictor device is significant for the efficient management of initial energy.

C. Solar-Harvesting Prediction Model

Each sensor node is powered by a rechargeable battery with a solar harvesting kit for the sensor network's maximum lifetime. Moreover, there are so many solar-harvesting issues affecting the harvesting rate of the sensor nodes, such as the intensity of sunlight, the angle of the sun, and the location of deployed sensor nodes. The prediction model's significance works based on the different parameters as a triple function (d, l, i). A day is divided into M nonoverlapping prediction intervals, and each slot length L consists of T duration, which means $(M.L.T = 24 \text{ h})$. However, we can identify the above triple function that d is day with l interval and refer to the particulars slot i. Fig. 1 has shown the solar-harvesting prediction model in EH-WSNs.

Furthermore, the description between photovoltaic (PV) cells and energy predictor devices is interrelated in solar-based harvesting systems. Because the output power of the PV cell depends on different factors, such as efficient irradiance value, load resistance, cell temperature, and solar incidence angle, these components are also affected by the other environmental conditions in EH-WSNs. Therefore, we can design a reliable solar-harvesting circuit by using an effective energy prediction technique. The PV cell can be simulated using the equivalent circuit. This lumped circuit incorporates a current generator to provide the short-circuit current I_{light}, and it is a function of solar irradiation. However, some important parameters are used to generate the light, incredibly reverse saturation current I_0, the series resistor R_S, and shunt resistor R_{SH}. Then, the coordination among solar cell module, temperature, and solar irradiance is given as follows:

$$I_{\text{solar cell}} = I_{\text{light}} - I_0 \left(e^{\frac{V_{\text{solar cell}} + I_{\text{solar cell}} R_S}{a}} - 1 \right) - \frac{V_{\text{solar cell}} + I_{\text{solar cell}} R_S}{R_{\text{SH}}} \quad (7)$$

where a is the modified solar cell ideal factor defined by

$$a = \frac{N_S \gamma k T_C}{q}. \quad (8)$$

Here, q is the electron charge value and k is Boltzmann's constant [23]. Some other values identified, such as γ is the general solar cell ideal factor, N_S is the number of solar cells in series, and T_C is based on cell temperature. In this article, we have assumed some essential considerations as follows.

1) All the sensor nodes are deployed in the outdoor-location x–y axis.
2) The typical deployment of the sensor node is random.
3) Each sensor node has a solar harvesting device with an efficient rechargeable battery.
4) This network is homogeneous based on the initial energy E_0 of nodes and computing power.
5) Here, the time slot is divided into the minutes of the day for the harvesting rate.
6) It is based on short-term (a few minutes to half an hour) as well as medium-term (a few hours) energy prediction.
7) The rate of EH is dynamic, and it depends on the seasonal to seasonal conditions.

III. Proposed Work

This section presents a profile-based energy prediction method called modified Pro-energy prediction in the solar-harvesting system. The working principle depends on the past day observation to derive efficient predictions on future energy intake. Prediction techniques concerning future energy availability also support minimizing the waste of energy resources. Furthermore, it is a prominent issue in the solar EH system because even if two sensor nodes are physically positioned, their harvesting rates significantly differ, affecting the size of storage devices. Generally, the size of the storage class is too tiny, especially supercapacitors and rechargeable batteries. Therefore, energy management is a complex task due to the size and time of EH-WSNs. One another approach can be possible to manage intraenergy resources using cluster formation in sensor networks. The cluster head (CH) coordinates the power control of their cluster members, and after gathering, all composite signal is sent to the BS through CH [24].

Associated with this kind of issue, our method can exhibit effective energy management solutions with the help of previous profile-based information. These profiles identify the harvested energy during a given number D of standard days. Once per time slot, the modified Pro-energy technique estimates the expected energy utility during the next time slot by looking at the cached profile that is the most similar to the current day. Furthermore, Fig. 2 shows the most similar profile of the following days, and here, five days are given to identify similarity. Second, the stored profile is elected among the typical ones, as it is most similar to the part

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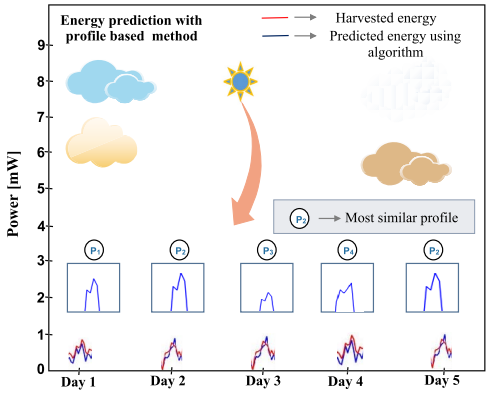

Fig. 2. Diagram of similar profile in modified Pro-energy technique.

of the present-day observed so far during the time slots of Day2. As the day progresses, the shape of the profile moves according to new observations. Three different profiles are employed for predictions during days 1, 3, and 4. Finally, current observations are selected again for the most similar profile day Day2.

The profile characteristic of the day is classified in different ways, such as sunny, cloudy, and rainy. However, the harvesting rate can be associated with any weather conditions. The modified Pro-energy method divides each day into the N equal-length time slots. Previous days harvested energy information is stored in matrix $(D \times N)$, reducing time and the memory overhead of the prediction approach. Besides, it is the most often used technique in prediction to divide equally distributed time slots. This is more beneficial to update future harvesting energy estimation only at the predefined instants at a particular time. Pro-energy works on the principle that solar energy reception at a given location follows a pattern. Therefore, our algorithm leverages the idea to look up the closest energy reception pattern match to the current day and build an accurate prediction. However, it comes up with an intelligent way of combining EH in the latest slot with the closest energy generation pattern.

The modified Pro-energy works the same as the Pro-energy method. Still, the only difference is estimating parallel error reduction based on the previous time slot of the day. The main drawback of the Pro-energy prediction technique is that it does not consider the error encountered during the prediction process. Our novel modified Pro-energy technique combines the existing algorithm with an error-moving average to enable the algorithm to self-correct and learn as the day progresses. Therefore, our approach would be more accurate with less prediction error margin compared to actual harvested energy in the following days. The energy profile contains different types of information, just like the weather condition of recent days. In some cases, the current sunny day has followed by cloudy weather after some time or vice versa. The modified Pro-energy method has also maintained one matrix $E(i, j)$ to keep the prior information pool of the D past days, each having N slots. Here, j is a sample slot within ith day of the previous time slots. However, there are three types of the subenergy matrix of day, and it contains an energy profile. It is fully based on the varying weather condition, such as E_{sunny}, E_{cloudy}, and E_{mix}. All three conditions calculate harvesting energy characteristics within the following cases as follows:

$$E'_{\text{sunny}} = \sum_{i=1}^{D} \sum_{j=1}^{N} E_{\text{sunny}}(i, j). \tag{9}$$

Equation (9) shows the energy prediction of a sunny day in different slots of the past day. Generally, the prediction error ratio (PER) of a sunny day is better, which means a minor error

$$E'_{\text{cloudy}} = \sum_{i=1}^{D} \sum_{j=1}^{N} E_{\text{cloudy}}(i, j). \tag{10}$$

The above equation has identified the prediction ratio of a cloudy day. Moreover, the error ratio of the cloudy day depends on the place to place

$$E'_{\text{mix}} = \sum_{i=1}^{D} \sum_{j=1}^{N} E_{\text{mix}}(i, j). \tag{11}$$

Equation (11) is based on the mixed day energy prediction, which means a sunny morning day followed by noon cloudy on a particular day and vice versa. This type of weather fluctuation of the day creates more prediction errors. The above terms are likewise called energy profile P of day.

However, modified Pro-energy identifies as the state-to-art term in solar-harvesting prediction and separately considers their behavior. All these separate parameters are recombined into a complete solar-harvesting prediction process. Therefore, the combination of all parameters is how much energy was harvested in the previous slot H, energy from most similar day θ and ρ introduced error encountered function of the time slots. Then the energy status to predict current-day energy information is defined as follows:

$$\widehat{E}(d, n) = \alpha H + (1 - \alpha)\theta - \rho(d, n). \tag{12}$$

Here, θ is a particularly significant issue of this approach, and it presents the most similar day in the different slots. The value of the α weight factor is identical to EWMA, WCMA, and Pro-energy. However, the MAE will estimate the energy level's similarity. The value of MAE is compared with the current day energy to elect the most similar day. However, the MAE function calculates the error between past days harvesting energy observation E_d, where d belongs to past days $(d_1, d_2, d_3, \ldots)$, and current days harvesting energy C in the respective time slots t

$$\text{MAE}(E_d, C) = \min_{E_d \in E} \sum_{i\,=\,t-K+1}^{t} \frac{1}{K} \left| C_i - E_{d_i} \right| \tag{13}$$

where K identifies the last K-slots to check similarity from the current. Moreover, the day's minimum MAE value is established as the most similar day, and this information is updated in the energy profile at the end of the day. This procedure uses multiple profiles of P to modify poor performance instead of a single profile. The numerous profiles identify fluctuating weather conditions during the harvesting process, such as a sunny morning day followed by a cloudy weather condition

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or a rainy afternoon, or vice versa. Each E is sorted based on the MAE in respective time slots of day, and the weight factor depends on the MAE. This brings us to change the term of the most similar day from θ with the weight profile (WP). This equation need to be computed to replace the value θ as WP

$$\mathrm{WP} = \frac{\sum_{j=0}^{P} W_j \cdot E_{dj}}{P-1} \tag{14}$$

where E_{dj} denotes the previous day's information based on their MAE. Let $E_{d1}, E_{d2}, E_{d3}, \ldots, E_{dP}$ is ordered list of the similarity profile with the current day C. However, W_j provides the weightage of similarity of each of the past P profile days to the current day, and it is estimated as follows:

$$W_j = 1 - \frac{\mathrm{MAE}(E_{dj}, C)}{\sum_{j=1}^{P} \mathrm{MAE}(E_{dj}, C)}. \tag{15}$$

From (15), we can identify that the lower the value of MAE, the higher the day's similarity with currents. However, to make them more accurate and robust for outliers, a value of the weight mean is very important in this method. The calculated mean value is collected for each day in the memory, thereby it can be observed with less error of days to be given more weight according to (15). One of the significant changes of this modified Pro-energy is identifying errors encountered prediction function of the previous time slots. Let the error observed on different days d within n slots be

$$\rho(d,n) = \beta\rho(d,n-1) + (1-\beta)\Big[\widehat{E}(d,n-1) - E(d,n-1)\Big] \tag{16}$$

where $\rho(d,n-1)$ is the expected error in our predicted energy $\widehat{E}(d,n)$, and the term $[\widehat{E}(d,n-1) - E(d,n-1)]$ denotes the actual error observed on the dth day of the $n-1$th slots. β is an exponential weighted average parameter to give us the error correction factor of the previous day in the different slots. Hence, the final equation of the modified Pro-energy with multiple profiles is

$$\widehat{E}(d,n) = \alpha H + (1-\alpha)\mathrm{WP} - \rho(d,n). \tag{17}$$

The above justification shows that our technique can predict the multiple slots' energy information with lower overheads and high accuracy. Therefore, we can accomplish initial power management in EH-WSNs. However, *Algorithm* 1 shows complete process.

Algorithm 1 Modified Pro-Energy Prediction

INPUT: $\alpha, \beta, D, E(i,j)$, and $thre$
OUTPUT: List of predictions of the given day D
Find the energy profile of different weather conditions using a threshold value to be adapted by seasonal variation.
Initialize $\alpha = 0.5, \beta = 0.2, D = 10$, and $thre = 0.4$.
for i: 0 to D **do**
  if $\sum_{j=0}^{N} E(i,j) > (1 - thre).E'$ **then**
    The ith day is sunny.
  else if $\sum_{j=0}^{N} E(i,j) < (1 + thre).E'$ **then**
    The ith day is cloudy.
  else
    The ith day is followed by sunny and cloudy condition.
  end if
end for
Calculate the $MAE(E_d, C)$ value using $Eq.(13)$.
for i: 0 to D **do**
  Assign the number of the days D.
  Calculate $WP = \frac{\sum_{j=0}^{P} W_j.E_{dj}}{P-1}$.
end for
Select most similar day based on $MAE(E_d, C)$ value.
Sort $(MAEs, key = index, order = desc)$.
for i: 0 to D **do**
  Calculate $W_j = 1 - \frac{MAE(E_{dj},C)}{\sum_{j=1}^{P} MAE(E_{dj},C)}$
  Identify parallel prediction error ratio using $Eq.(16)$.
  Define $\rho = 0$.
  for s: 0 to slot **do**
    $\rho = \beta \times \rho + (1-\beta) \times \big[\widehat{E}(d,n-1) - E(d,n-1)\big]$
    $\theta = E^T[slot][D-1:0] \times W$.
    $\widehat{E}[d,n] \leftarrow \alpha \times E(d,n-1) + (1-\alpha) \times \theta - \rho$.
  end for
end for

Lemma 1: The time complexity of modified Pro-energy prediction algorithm is $\mathcal{O}(D \times N)$ in each case.

Proof: The time complexity of modified Pro-energy depends on various factors, especially energy profile, calculation of MAE, and parallel error correction of past slots. The energy profile consists of weather conditions, such as day, cloudy, and a mix of past days, and it takes for execution $\mathcal{O}(D)$. However, the calculation of the MAE value to find the most similar days from previous days is $\mathcal{O}(D)$. Similarly, the parallel error correction function of the previous day in respective time slots takes $\mathcal{O}(D \times N)$. Then, total time complexity is $\mathcal{O}(D) + \mathcal{O}(D) + \mathcal{O}(D \times N) = \mathcal{O}(D \times N)$.

Remark 1: Note that the harvesting system depends on different environmental monitoring types, and the prediction of harvesting energy is variable. The EH-rate is fully based on the seasonal condition at a certain time slot of the day D_t. There are two instants used to identify of EH-rate in the different condition, such as $h_r \in P_{\max}$ and $h_r \in P_{\min}$. According to the modified Pro-energy prediction technique, the condition is followed for certain minutes of the day. The rule of the harvesting system theory, if the weather condition is sunlight day D_t, then harvesting rate increases $D_{\max}(h_r^+)$; otherwise decreases $D_{\max}(h_r^-)$.

Remark 2: One of the strong justification of the harvesting system is that if the ratio of prediction energy is elegant, then it increases the energy level E_L of the battery $B_i(t)$. This is very practical because the energy predictor device is directly connected to the sensor node battery in the solar-harvesting device. The residual energy entirely depends on the solar light intensity, and it also varies in different time slots of the day. However, the prediction technique's role is significant because it identifies how much energy would come next time slot of the day. The conversion rate of the rechargeable battery's residual power is associated with the weather condition in the harvesting system. Therefore, the importance of efficient prediction is needed for initial energy management.

IV. Result Analysis of the Modified Pro-Energy

The experimental evaluation of the modified Pro-energy prediction technique has been implemented in the MATLAB 2016 environment. Moreover, we have divided our

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TABLE I
IRRADIANCE

Location	Energy
Global PSP	W/m^2
Direct NIP	W/m^2
Diffuse PSP (sband corr)	W/m^2
Reflected PSP	W/m^2
Global 90-North	W/m^2
Global 90-North (gnd shielded)	W/m^2
Global 90-North (gnd reflected)	W/m^2
Global 90-East	W/m^2
Global 90-South	W/m^2
Global 90-South (gnd shielded)	W/m^2
Global 90-South (gnd reflected)	W/m^2
Global 90-West	W/m^2

implementation into four portions, such as different datasets collection, evaluation of the proposed algorithm, error metric analysis, and energy level (E_L) prediction of the sensor networks.

A. Experimental Setup

For the simulation setup, we consider a fixed weighting factor α for prediction $\alpha = 0.5$ and $\beta = 0.2$. The dataset is taken from the National Renewable Energy Laboratory (NREL)[1] and a description of all data fields collected from different instruments is mentioned in the NREL repository. Moreover, this dataset is a collection of real-life solar-based data widely used for practical implementation in EH-WSNs. Besides, the irradiance solar value of various locations is presented in Table I, and a subset of the dataset is illustrated to justify actual harvesting energy in Table II.

B. Evaluation of the Proposed Algorithm

The proposed algorithm is pitted against EWMA, WCMA, and Pro-energy to establish merit and compare the results. Moreover, the modified Pro-energy performance compares adequately with the ground-truth solution. Fig. 3 shows that our proposed algorithm consistently performs better results than the previous works. This result provides a much smoother curve with minor prediction differences. It is trivial to verify that our intuition (the addition of errors generated during previous mistakes) helps the prediction signal perform better and better as the day progresses. As expected, WCMA outperforms EWMA by considering the offsets observed by the weather changes in the present day. In contrast, Pro-energy defeats the two moving averages in both situations of high variance (bottom) and low variance (top). It is also easy to remark that the proposed approach performs betters than Pro-energy, resulting in a much closer approximation of the ground-truth signal. Fig. 3(a) has determined the comparative analysis among ground truth, Pro-energy, and the proposed technique. Moreover, Fig. 3(b) and (c) also justify comparison among the existence prediction methods.

[1]http://dx.doi.org/10.5439/1052221 (NREL: Measurement and Instrumentation Data Center)

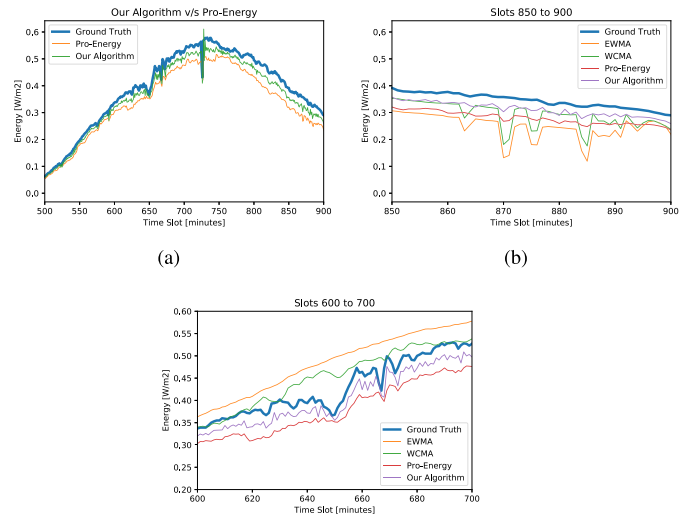

Fig. 3. Performance analysis. (a) Pro-energy versus proposed algorithm on Day 10. (b) Comparison of algorithms for smooth change. (c) Comparison of algorithms highly varying in data points.

C. Error Metric

We estimated the error performance of modified Pro-energy by comparing the amount of actual energy harvested during each time slot. However, the error metric is evaluated using mean absolute prediction percentage error with N slots is defined as follows:

$$\text{Error}[\%] = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{H_i - \widehat{E}_i}{H_i}\right| \tag{18}$$

where H_i is the actual energy and $\widehat{E}_i$ is our estimated energy, averaged over all data points. We select four random days from the year and look at the Error[%] values. The trends follow our expectations, and modified Pro-energy outperforms all other methods by a significant margin. EWMA prediction technique has no mechanism to check the weather conditions due to the lack of storage class of the last energy information and current-day, so it produces high errors in temporary changes. However, we can see in Fig. 4(a) and (b). Fig. 4(a) and (b) justifies the error based on different weather, changing sunny and cloudy. Here, Fig. 4 shows a huge variation in the EMWA prediction region, only weather changing from sunny to cloudy. The value of α is a particularly significant issue with seasonal changes in the harvesting system. Pro-energy prediction gives better performance than the EWMA and WCMA prediction methods. Similar things are happening in respective Fig. 4(c) and (d).

Our proposed scheme estimates the energy generation of the days with the most similar energy generation profile. Finding the most similar energy profile of the day has been calculated to compare the average prediction errors of the previous days. Therefore, the most similar day has the lowest absolute prediction error based on the last Pro-energy. Sometimes, it may also be the wrong experience based on changing the weather. Fig. 4(c) and (d) are identified as an average PER of all several prediction techniques, but we have realized that our algorithm has a minor error. Moreover, Table III is part of the

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TABLE II
SUBSET OF THE DATASET

Sl. No	DATE (MM/DD/YYYY)	MST	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$	Global CMP22 (vent/cor) $[W/m^2]$
0	1/26/2019	07:16	3.97621	4.44060	4.27966	4.83216	8.42773	4.71733	5.19894	-0.001	0.000	1.98487	0.000
1	1/26/2019	07:17	4.96652	5.46627	5.29066	5.97006	5.57426	5.95183	6.60461	-0.001	0.000	2.64293	0.000
2	1/26/2019	07:18	5.75069	6.28833	6.13483	6.76291	10.39120	6.70633	7.65581	-0.001	0.000	2.64293	0.000
3	1/26/2019	07:19	5.33213	5.86996	5.74418	6.39497	10.08150	6.28613	6.44981	-0.001	0.000	3.01208	0.000
4	1/26/2019	07:20	5.09743	5.63157	5.47730	6.20357	10.13100	6.13155	5.68789	-0.001	0.001	3.33309	0.001
5	1/26/2019	07:21	5.52233	6.09820	5.91677	6.68798	10.69720	6.60658	6.30313	-0.001	0.001	3.83597	0.000
6	1/26/2019	07:22	6.23630	6.84322	6.67704	7.41612	11.46370	7.32754	7.47250	0.000	0.001	4.26931	0.000
7	1/26/2019	07:23	7.13014	7.77046	7.60138	8.30298	12.33090	8.21539	8.73559	0.000	0.002	4.72939	-0.001
8	1/26/2019	07:24	8.33020	8.93881	8.80785	9.29760	13.31010	8.95392	9.37937	0.000	0.002	5.46233	-0.001
9	1/26/2019	07:25	10.18070	10.73690	10.50250	10.41910	14.83740	9.94323	11.63660	0.000	0.002	7.24920	0.000
10	1/26/2019	07:26	13.68170	14.13010	13.63650	11.72600	17.14640	11.12610	14.14230	0.000	0.002	10.53949	0.000
11	1/26/2019	07:27	16.43840	16.67980	16.04660	12.55810	18.87470	11.966700	16.55840	0.000	0.003	12.83990	-0.001
12	1/26/2019	07:28	14.25870	14.59730	12.92810	12.92810	18.00440	12.47980	15.05500	0.001	0.003	10.26120	0.000
13	1/26/2019	07:29	13.41660	13.69800	13.69810	13.85820	18.09850	13.55450	15.62540	0.001	0.004	8.83910	0.000
14	1/26/2019	07:30	15.02990	15.5330	15.33630	15.19730	19.50140	14.90770	17.38150	0.001	0.004	9.99368	0.000
15	1/26/2019	07:31	16.12210	17.68650	16.42640	16.40060	20.80310	16.18890	17.84190	0.001	0.004	10.77480	0.000
16	1/26/2019	07:32	16.93940	17.53810	17.33180	17.51270	21.81660	17.32260	18.86060	0.002	0.005	11.07430	0.000
17	1/26/2019	07:33	17.90580	18.53060	18.29950	18.58580	22.94000	18.44710	20.11550	0.002	0.005	11.62530	0.000

TABLE III
ERROR ANALYSIS OF DIFFERENT DAYS ERROR [%]

No. of Days	EWMA	WCMA	Pro-Energy	Our Algorithm
Day 10	1.254554	1.0993933	0.933787	0.884650
Day 30	1.837535	1.787515	2.088974	1.069490
Day 60	1.106148	0.983891	0.975873	0.714068
Day 100	3.020421	1.364042	0.931500	0.917892

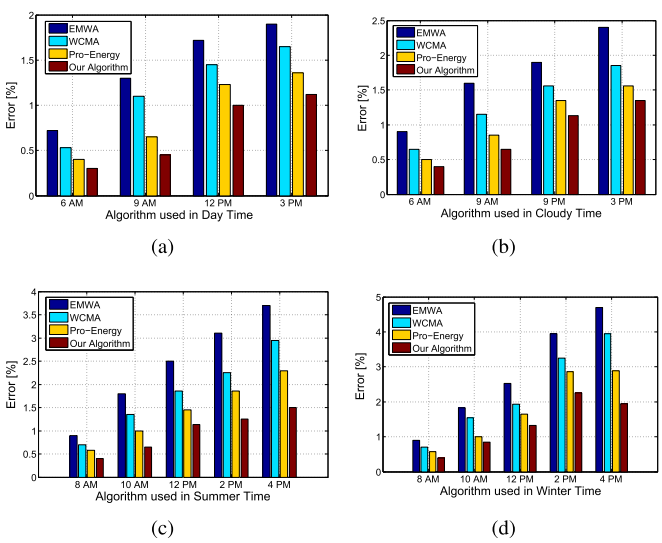

Fig. 4. Error metric with (a) average prediction error ratios in the daytime, (b) average prediction error ratios in cloudy time, (c) error prediction with summer time slot, and (d) error prediction in winter.

TABLE IV
PREDICTION ERROR RATIOS OF DIFFERENT TECHNIQUES

Algorithms	Error [%]
EWMA	1.2545536136392341
WCMA	1.093933399644929
Pro-Energy	0.9337872464759861
Our Algorithm	**0.8846498481024798**

different days' error analysis, and Table IV is an important term in the simulation work to show the prediction error accuracy of different techniques.

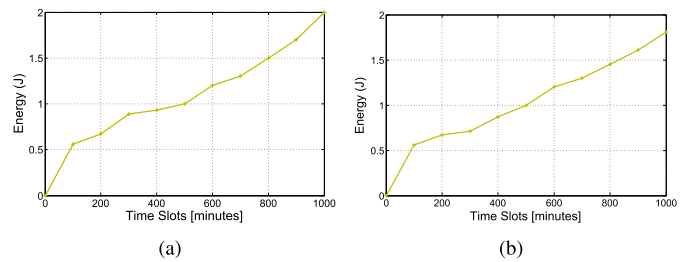

Fig. 5. Analysis of E_L. (a) Energy level of the sensor networks on a sunny day. (b) Energy level of the sensor networks on a cloudy day.

We validate our modified Pro-energy prediction technique by running the experiment on a dataset for 365 days and averaging over all days. According to PER analysis, EWMA generates a 16.1% more accurate error of ten days compared to WCMA. Similarly, Pro-energy gives less error prediction than WCMA by 16%. Finally, we notice that the modified Pro-energy technique outperforms Pro-energy by 4.9% and WCMA by 20.9%, which is a significant gain in accuracy with little computation overhead. This gain in accuracy helps optimize the battery life of various sensor nodes.

D. Energy Level E_L of Sensor Networks

The EH theory defines that if the PER is less, then the energy level E_L of the battery will be high. Different factors are responsible for charging the battery, specifically different weather conditions, less error prediction accuracy, quality of the rechargeable battery, and location of the sensor node. This theory is usually dynamic because the solar-harvesting system affects the battery level of the sensor node. According to the above condition, Fig. 5(a) and (b) justifies the energy level E_L of the sensor node. Fig. 5(a) and (b) shows that the energy level increases with time, irrespective of sunny and cloudy days.

E. Residual Energy Measurement

The performance of the prediction method depends on the error factor. If the error ratio is less, network performance would be high, affecting the battery's residual energy, network lifetime, and throughput. Concerning these issues, we present

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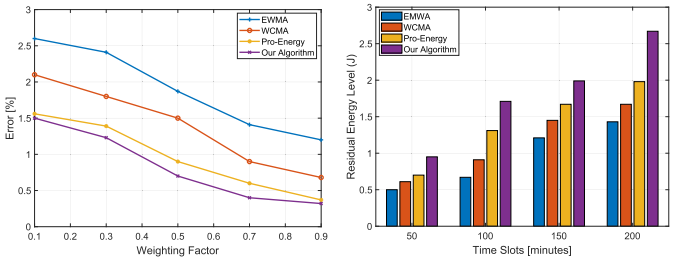

(a) (b)

Fig. 6. Performance of the network. (a) Error analysis with various weighting factors α. (b) Residual energy level of different prediction techniques.

the performance of various weighing factors and residual energy prediction algorithms. Furthermore, Fig. 6(a) justified error margin analysis of various weighting factors α, and corresponding to this validation, Fig. 6(b) shows the residual energy performance of algorithms.

V. Conclusion

The sensor network application has expanded in recent times for several regions. One of the sensor network's significant issues is energy-saving and replacing the sensor node battery. However, it is too tricky in the unreachable area. The researcher has solved the limitation of energy resources using the EH technique. In this article, we have proposed a solar EH prediction in EH-WSNs. Energy prediction is an initial procedure to manage the ambient resource. The prediction approach is entirely based on the weather condition because of what amount of energy will come in the next time slots of the day. This type of argument does not always give exact results but using prediction techniques, we can estimate based on previous day energy information in the respective time slots. Related to the prediction issue, the modified Pro-energy technique gives better results and beats other methods, especially EWMA, WCMA, and Pro-energy. The prediction accuracy of this method is 4.9%. However, it helps optimize the sensor node battery by gaining efficiency and prolonging battery life in EH-WSNs.

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